

k-means Core Concepts (1–10)

- 1. Q:** What is the fundamental goal of clustering in unsupervised machine learning?
A: To group similar data points together and separate dissimilar points without using labeled data.
 - 2. Q:** How do points within a cluster compare to each other, and between clusters?
A: Points within a cluster are similar; points in different clusters are dissimilar.
 - 3. Q:** What does the “k” represent in k-means clustering?
A: The number of clusters to form.
 - 4. Q:** What is the term for the center of a cluster?
A: Cluster centroid (or cluster mean).
 - 5. Q:** Which distance metric is typically minimized in k-means?
A: Squared Euclidean distance.
 - 6. Q:** What does WCSS stand for, and what is the goal regarding it?
A: WCSS = Within-Cluster Sum of Squares; the goal is to minimize it.
 - 7. Q:** Write the formula for WCSS.
A:
$$\text{WCSS} = \sum_{j=1}^k \sum_{\vec{x}_i \in C_j} |\vec{x}_i - \vec{m}_j|^2$$
 - 8. Q:** If a point is assigned to a cluster, what must the centroid satisfy?
A: The centroid must be the closest one to the point among all cluster centroids.
 - 9. Q:** What is the minimum possible value of (k)?
A: 1
 - 10. Q:** If (k) is unknown, how can it be estimated?
A: Using the Elbow Method.
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II. The k-means Algorithm Steps (11–20)

- 11. Q:** What is the first step in the k-means algorithm?
A: Initialize (k) cluster centroids randomly or by heuristic.
- 12. Q:** What are the two main initialization methods?
A: 1) Random selection of (k) points as centroids. 2) Random assignment of points to clusters and compute initial centroids.

13. Q: Describe the Assignment step.

A: Assign each point to the nearest cluster centroid.

14. Q: Write the condition for assigning a point to a cluster.

A: $(\vec{x}_i \in C_j^{(k+1)})$ if $(\|\vec{x}_i - \vec{m}_j^{(k)}\|^2 \leq \|\vec{x}_i - \vec{m}_l^{(k)}\|^2)$ for all (l) .

15. Q: What is the goal of the Mean Update step?

A: Recalculate each cluster mean as the average of points assigned to it.

16. Q: Formula for updated cluster mean?

A: $(\vec{m}_j^{(k+1)}) = \frac{1}{|C_j^{(k+1)}|} \sum_{\vec{x}_i \in C_j^{(k+1)}} \vec{x}_i$

17. Q: What statistical term does the Mean Update compute?

A: Arithmetic mean (centroid).

18. Q: When does k-means terminate?

A: When cluster assignments no longer change or WCSS change is minimal.

19. Q: If assignments change, what happens next?

A: Recompute cluster means and repeat the assignment step.

20. Q: Why can initial centroids affect the final result?

A: Different initial centroids can lead to different **local minima**, affecting final clusters.

III. Mathematical and Practical Aspects (21–30)

21. Q: Constraint on (k) given (n) points?

A: $(1 \leq k \leq n)$

22. Q: What characteristic of cluster means does k-means improve iteratively?

A: Minimizes intra-cluster variance (makes points closer to their centroids).

23. Q: Symbolic expression for squared Euclidean distance?

A: $(\|\vec{x}_i - \vec{m}_j\|^2)$

24. Q: If a cluster has only one point, what is its mean?

A: The mean equals that single point.

25. Q: Ultimate goal of minimizing WCSS?

A: Make points within each cluster as similar as possible, reducing variance.

26. Q: How does similarity relate to WCSS?

A: Minimizing WCSS ensures points in a cluster are highly similar.

27. Q: What does WCSS stabilization indicate?

A: The algorithm has converged.

28. Q: If Manhattan distance is used, what is the cluster center called?

A: Median (instead of mean).

29. Q: How are initial means computed if points are randomly assigned to clusters?

A: Compute the **arithmetic mean** of points in each cluster.

30. Q: K-means converges to what type of minimum?

A: A **local minimum** of WCSS (not necessarily global).